

Mapped CO(s) Number	Practical/ Lab Session Outcomes (LSOs)	Laboratory Experiment/ Practical Titles
CO1	<i>LSO 1.1</i> Measure voltage, current, and resistance using standard instruments. <i>LSO 1.2</i> Apply color code to identify resistor values. <i>LSO 1.3</i> Compare expected with measured values.	LE- 1 Measurement of various electrical parameters
CO1	<i>LSO 2.1</i> Verify ohm's law in a nonlinear circuit	LE- 2 Verification of ohm's law in a nonlinear circuit
CO1	<i>LSO 3.1</i> Apply KVL to a series circuit and verify the sum of voltage drops equals applied voltage. <i>LSO 3.2</i> Apply KCL to a parallel circuit and verify the sum of branch currents equals total current.	LE- 3 Verification of Kirchhoff's Current Law and Voltage Law
CO2	<i>LSO 4.1</i> Measure average value, peak value and RMS value of AC waveform. <i>LSO 4.2</i> Measure time and frequency of AC waveform using CRO	LE- 4 Measurement of AC signal parameters
CO2	<i>LSO 5.1</i> Measure impedance, current, and power factor in a series RL and RC circuit.	LE- 5 Measurement of impedance, current, and power factor
CO2	<i>LSO 6.1</i> Measure power and determine pf in a circuit connected to a resistive and inductive load <i>LSO 6.2</i> Differentiate between the power and PF obtained in resistive and inductive circuit	LE- 6 Measurement of power and power factor in a single-phase ac series in a - (i) Resistive circuit (ii) Inductive circuit.
CO2	<i>LSO 7.1</i> Connecting different types of lamps <i>LSO 7.2</i> Observe the power consumption and energy saved due to LED lamp	LE- 7 Energy efficient lamp - Measurement of power consumption of an Incandescent lamp, fluorescent lamp (tube light) and a LED light of same rating.
CO2	<i>LSO 8.1</i> Make a star and delta connection to measure line and phase voltage in a 3 phase AC circuit.	LE- 8 Line and Phase voltage measurement of star - delta connection circuit in a 3 phase AC circuit
CO4	<i>LSO 9.1</i> Determine the voltage ratio and polarity of a given single-phase transformer <i>LSO 9.1</i> Interpret the results.	LE- 9 Determination of (i) Voltage ratio (ii) Polarity of a single-phase Transformer
CO4	<i>LSO 10.1</i> Identify the correct specification autotransformer to be used according to load current	LE- 10 Autotransformer specification
CO4	<i>LSO 11.1</i> Vary input supply voltage using a single-phase transformer <i>LSO 11.4</i> Vary input ac supply voltage using a single-phase autotransformer.	LE- 11 Variation of input supply voltage using a single-phase autotransformer

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CO4	<i>LSO 12.1</i> Identify a transformer by its ratings. <i>LSO 12.2</i> Determine the voltage ratio and polarity of a given single-phase transformer. <i>LSO 12.3</i> Measure primary and secondary voltages under no-load and load conditions.	LE- 12 Determination of the voltage ratio and polarity of a single-phase transformer.
CO4	<i>LSO 13.1</i> Determine losses, efficiency and voltage regulation of a single-phase transformer <i>LSO 13.2</i> Interpret the results	LE- 13 Determination of the performance of a single-phase transformer by direct load test
CO4	<i>LSO 14.1</i> Determine losses, efficiency and voltage regulation of a single-phase transformer using OC and SC test <i>LSO 14.2</i> Interpret the results	LE- 14 Determination of the performance of a single-phase transformer OC and SC test
CO5	<i>LSO 15.1</i> Identify the parts of a three-phase induction motor <i>LSO 15.2</i> Operate a three-phase induction motor using DOL starter.	LE- 15 Operation of induction motor and starter.
CO5	<i>LSO 16.1</i> Demonstrate the cut section model of a given type of motor Determine efficiency under varying load.	LE- 16 Demonstration of cut-out sections of machines: DC machine, three phase induction machine, Single phase induction machine and synchronous machine.
CO5	<i>LSO 17.1</i> Change the phase sequence of a given 3 phase induction motor. <i>LSO 17.2</i> Observe the reversal of direction of the motor	LE- 17 Reverse the direction of a given three phase induction motor by reversing the Phase sequence
CO5	<i>LSO 18.1</i> Identify different parts of a single-phase induction motor <i>LSO 18.2</i> Reverse the direction of rotation of a single-phase induction motor of a ceiling fan, single phase induction motor and universal motor.	LE- 18 Identification of different parts of a single-phase induction motor and reversing the direction of rotation